Lecture 07

Policymaking Under Uncertainty

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AEM 4510

Roadmap

1. Motivating examples of environmental uncertainty
2. Weitzman: taxes versus standards under MD and MAC uncertainty
3. Manne–Richel: abatement today under uncertainty about the future

Uncertainty and the environment

Learning about arctic sea ice



Uncertainty in marginal damages

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Climate change

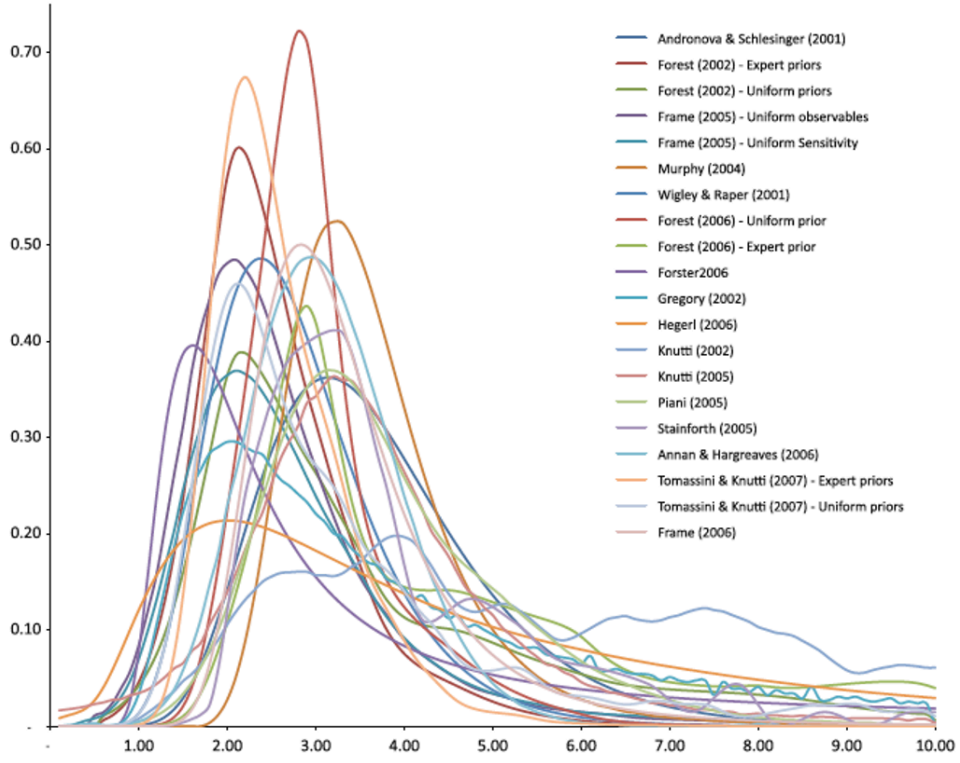
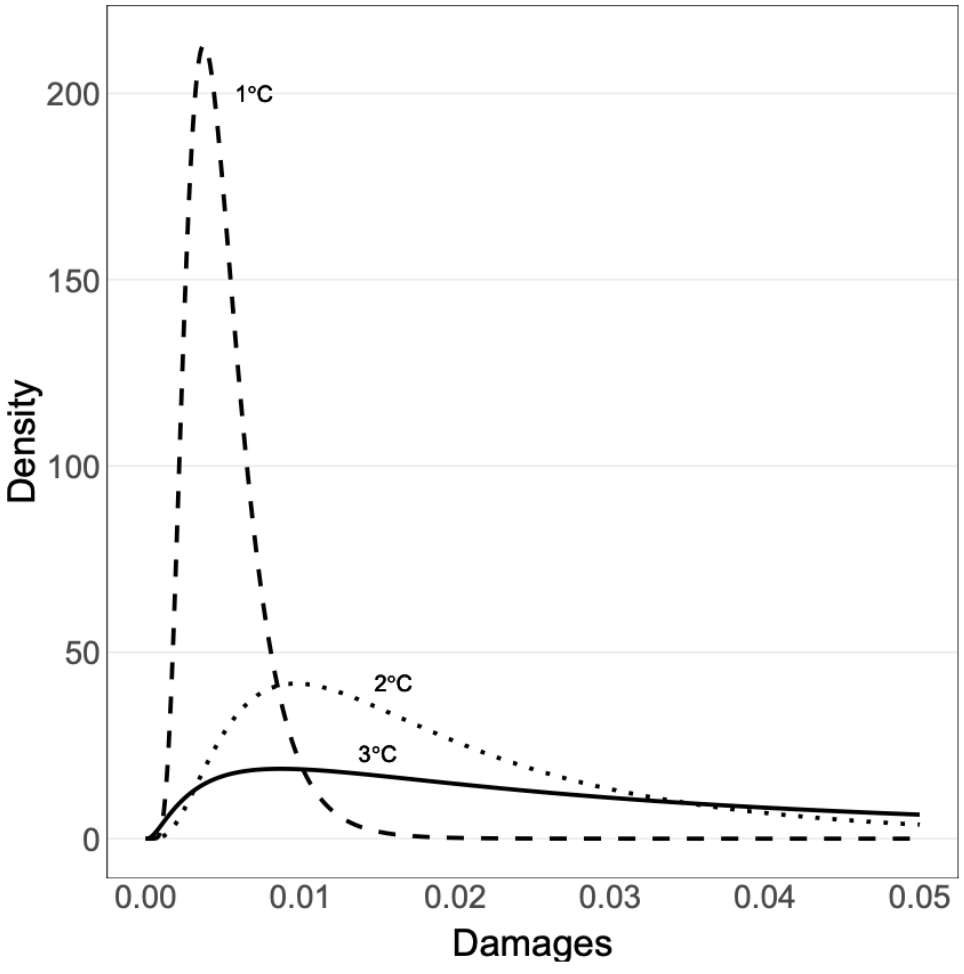


Fig. 1 Estimated probability density functions for climate sensitivity from a variety of published studies, collated by Meinshausen et al. (2009)

These are our best estimates of the distribution of the **equilibrium climate sensitivity:** how much the earth warms on average if we double CO_2 concentrations

Uncertainty in ECS means we are uncertain about the true costs of emitting a ton of CO_2 (the social cost of CO_2)

Climate change



There's also uncertainty over how costly a given amount of warming will be

- How bad is 1 degree of warming for agricultural yields?

These are distributions of damages to global GDP at given levels of warming

Rudik (2020)

Uncertainty

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- Uncertainty in marginal damages (MD)

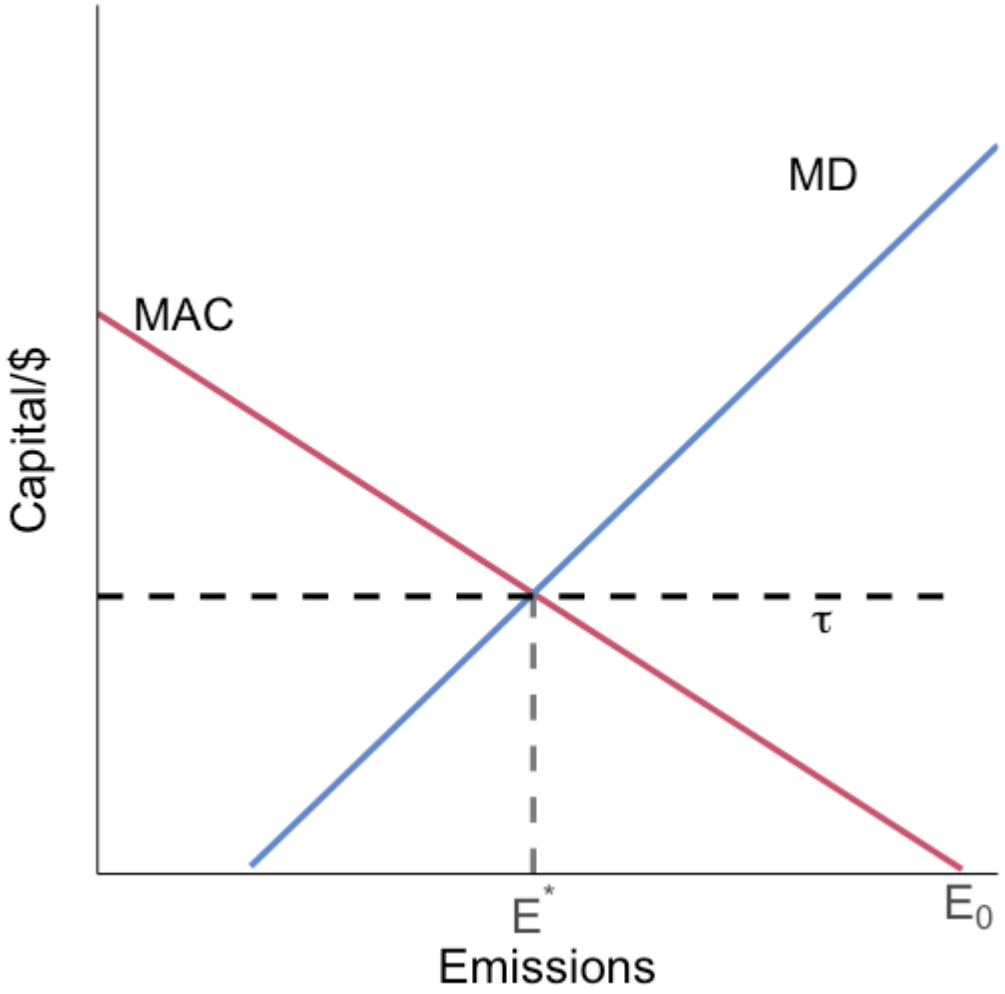
Uncertainty

Uncertainty can factor into environmental policy-making in a number of ways:

- Uncertainty in marginal damages (MD)
- Uncertainty in marginal abatement costs (MAC)

The type of uncertainty can have implications for instrument choice

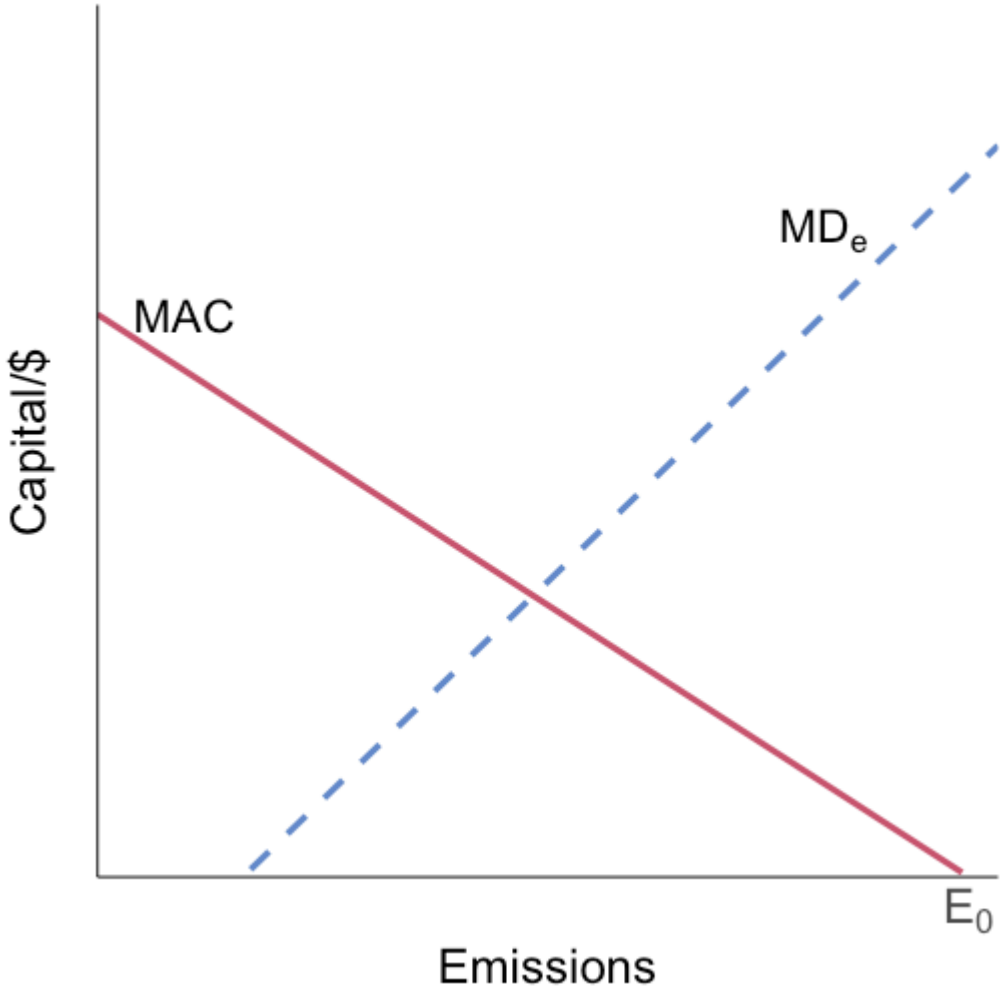
Policy under uncertainty



With **perfect information** we can achieve the efficient outcome using either a tax τ or an emission standard $\bar{E} = E^*$

Will this hold true even with uncertainty about MD or MAC?

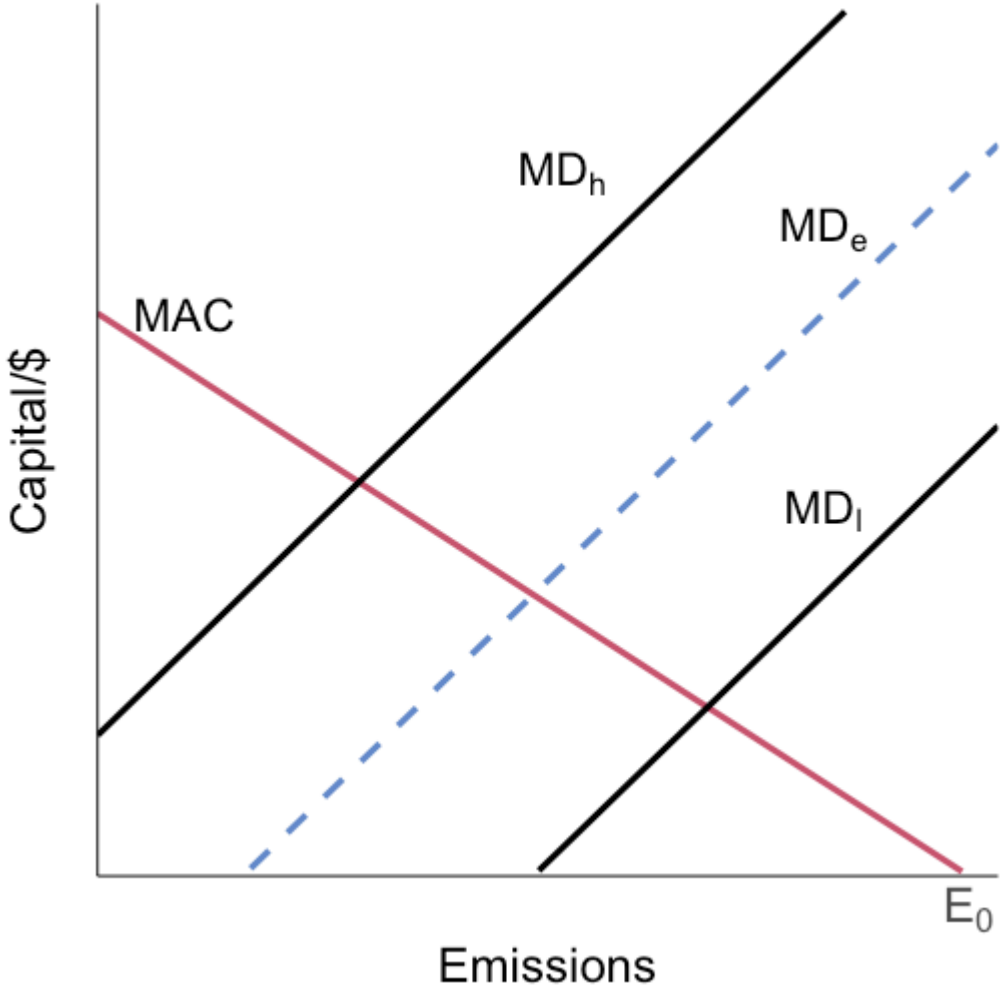
Policy under MD uncertainty



If we are uncertain about MD how should we set policy?

Suppose we have a best estimate, MD_e

Policy under MD uncertainty

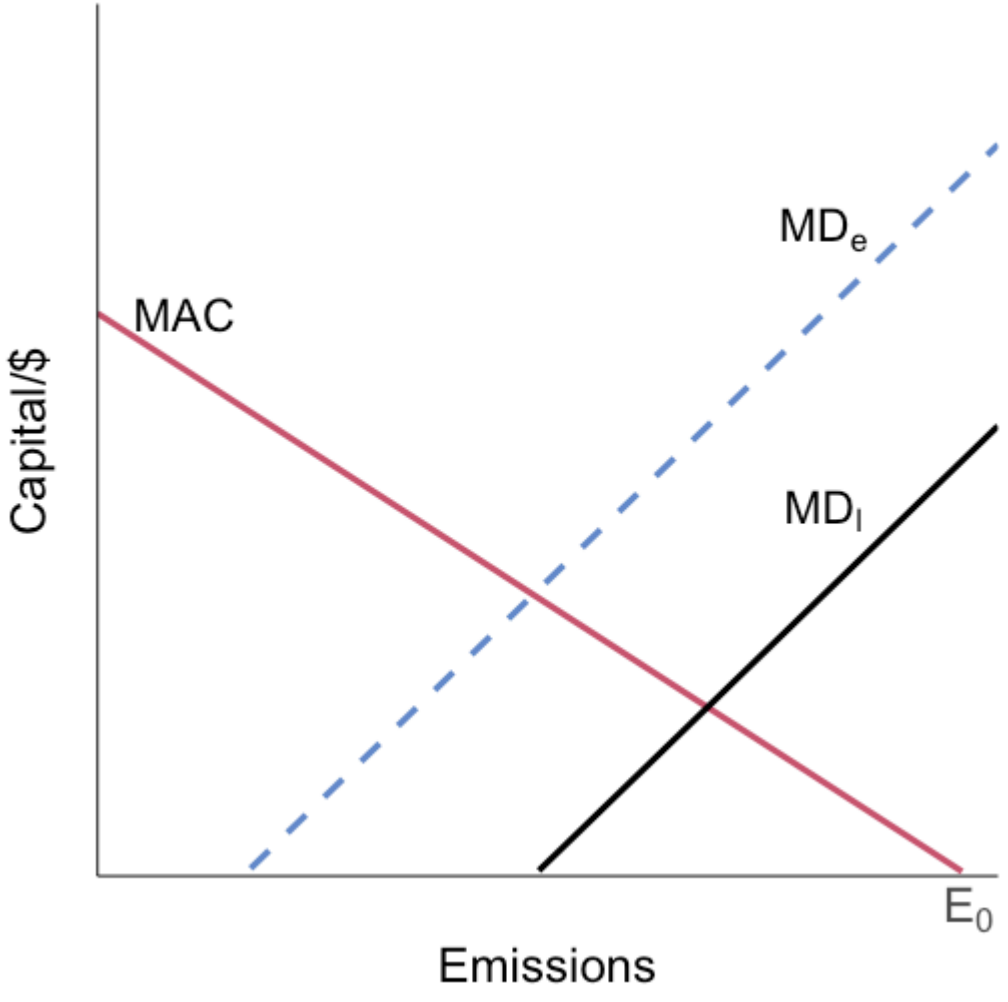


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Unfortunately our estimate was wrong, and the true MD curve is given by MD_h or MD_l

Policy under MD uncertainty



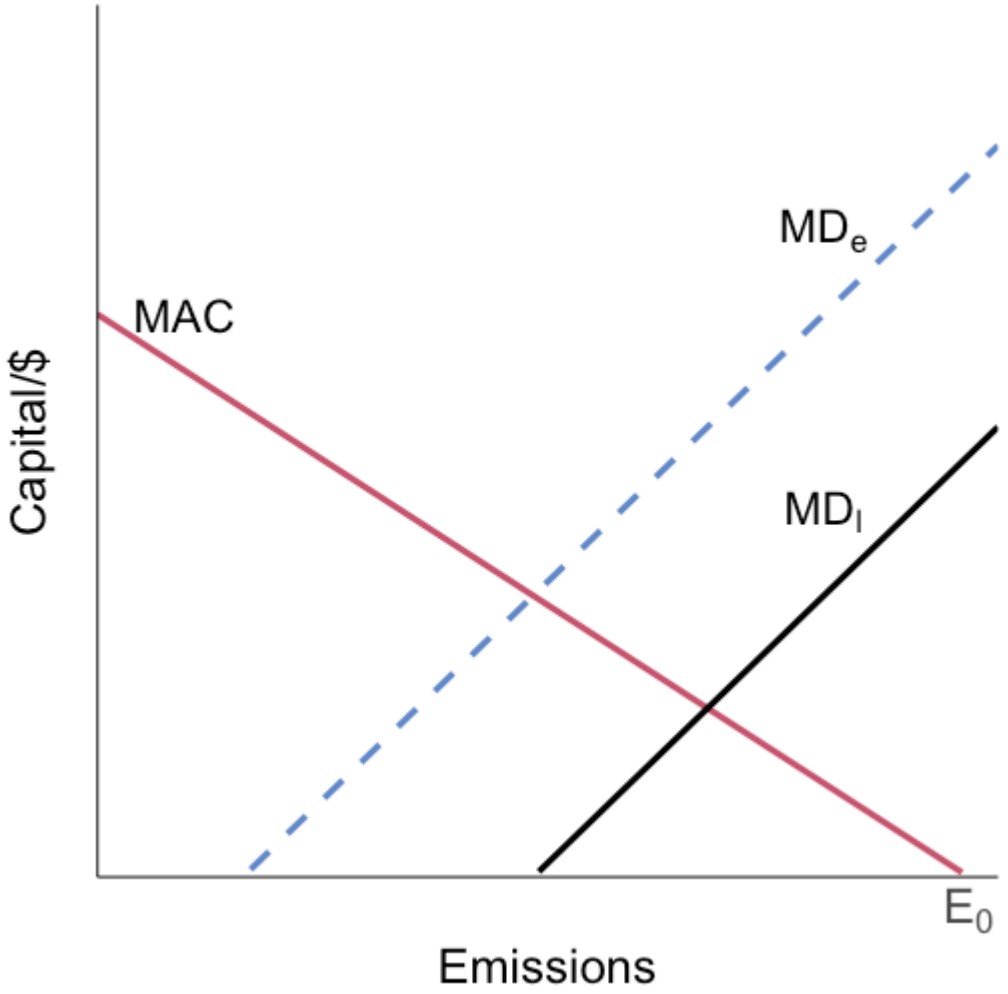
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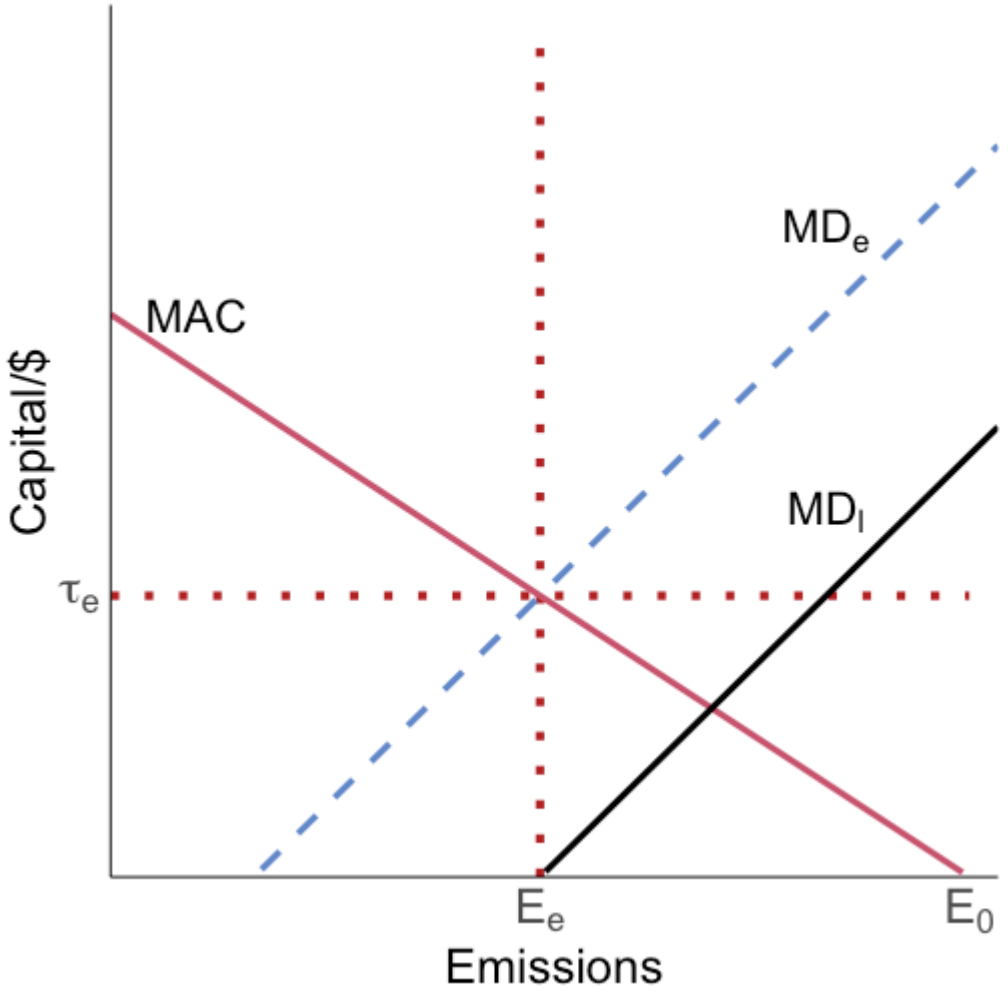
Let's say the actual MD is MD_l, what happens under our policies?

Policy under MD uncertainty



First: what tax and standard does the regulator set?

Policy under MD uncertainty



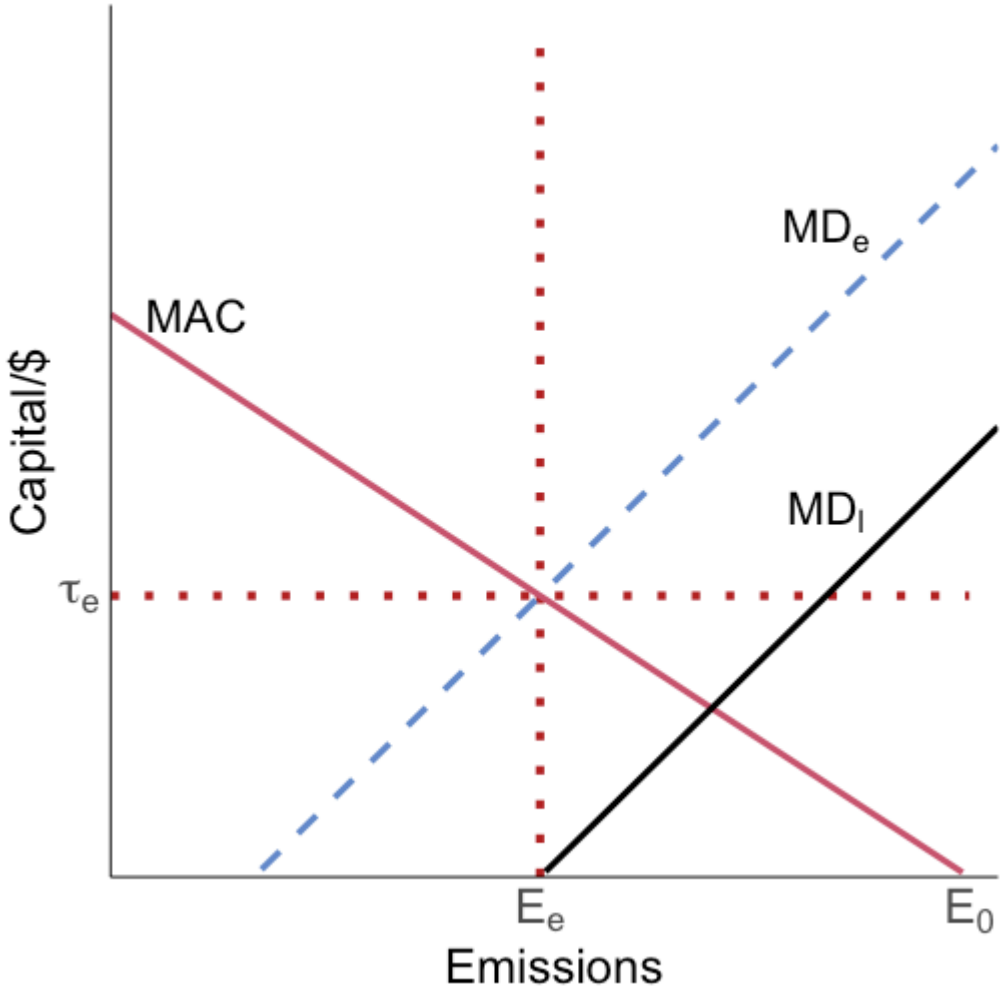
First: what tax and standard does the regulator set?

The regulator selects a standard where she thinks MAC = MD: E_e

The regulator selects a tax where she thinks MAC = MD: τ_e

With these two policies in place, what does the firm do?

Policy under MD uncertainty



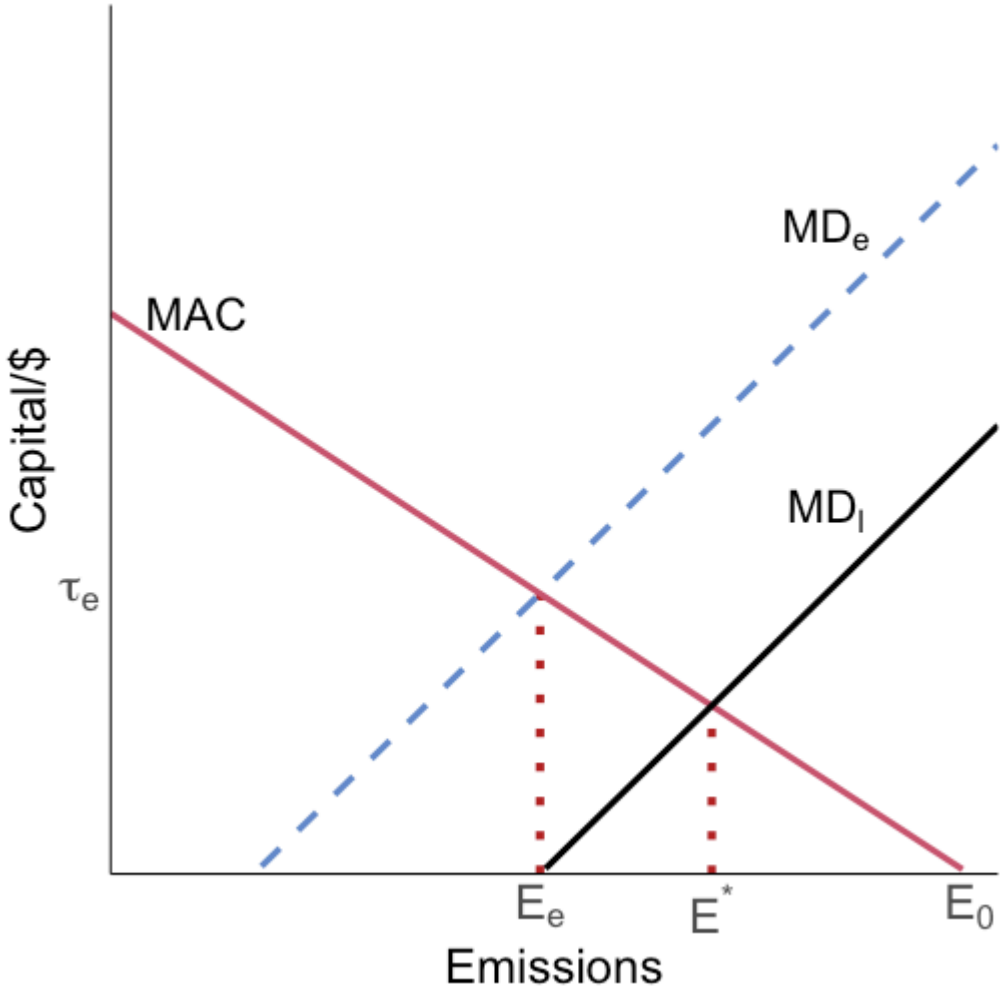
The firm emits at E_e under both the tax and standard

We get the same outcome

Is there any deadweight loss?

If so, how much?

Policy under MD uncertainty

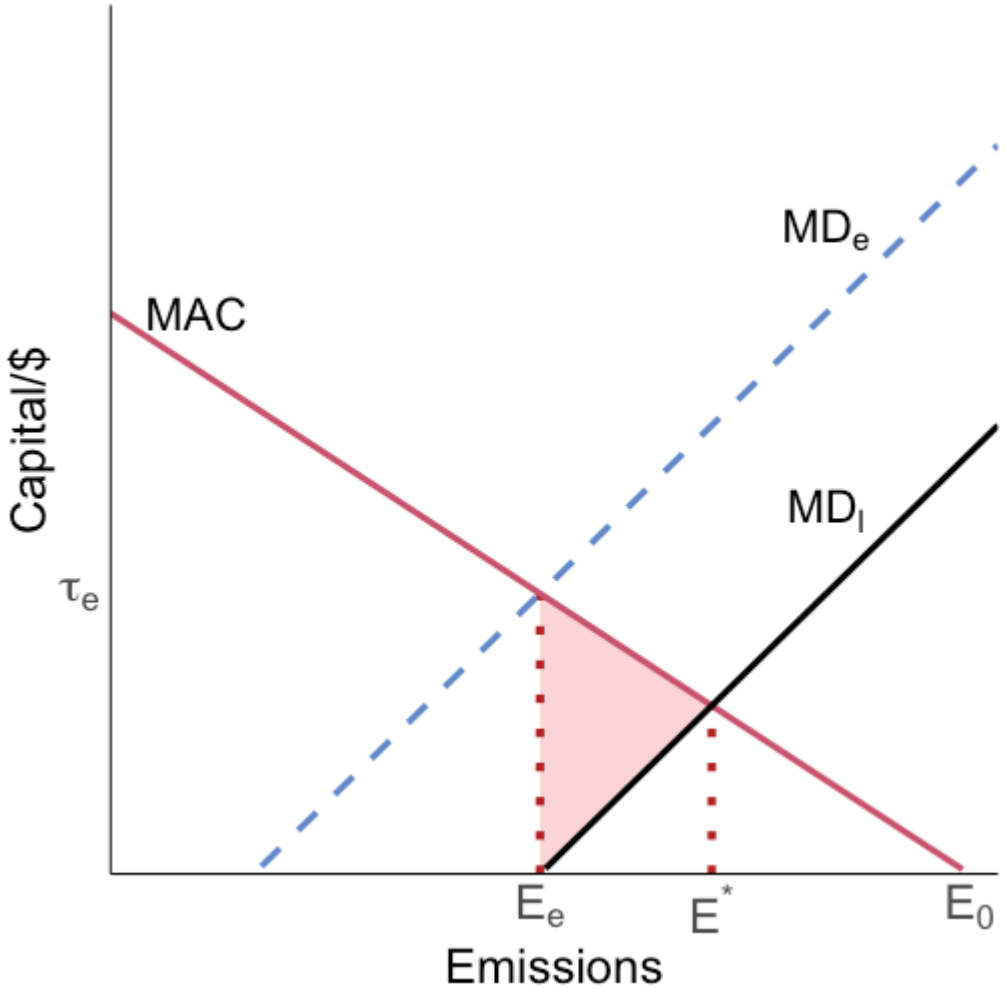


We actually want to be at E^*

We are emitting too little because we overestimated damages

What is the DWL associated with under-emission?

Policy under MD uncertainty

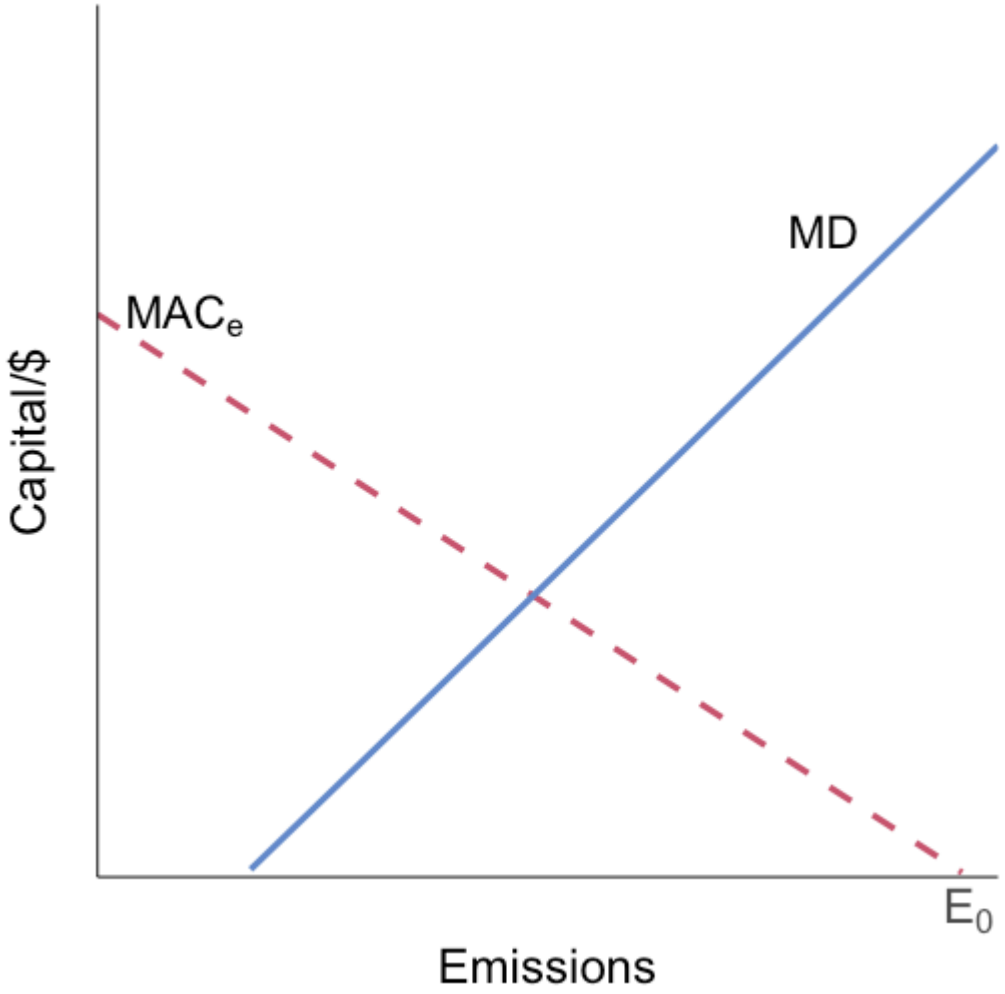


DWL comes from abatement costs in excess of the avoided damages

This is the area in red

With uncertain MD, we do not get the efficient outcome!

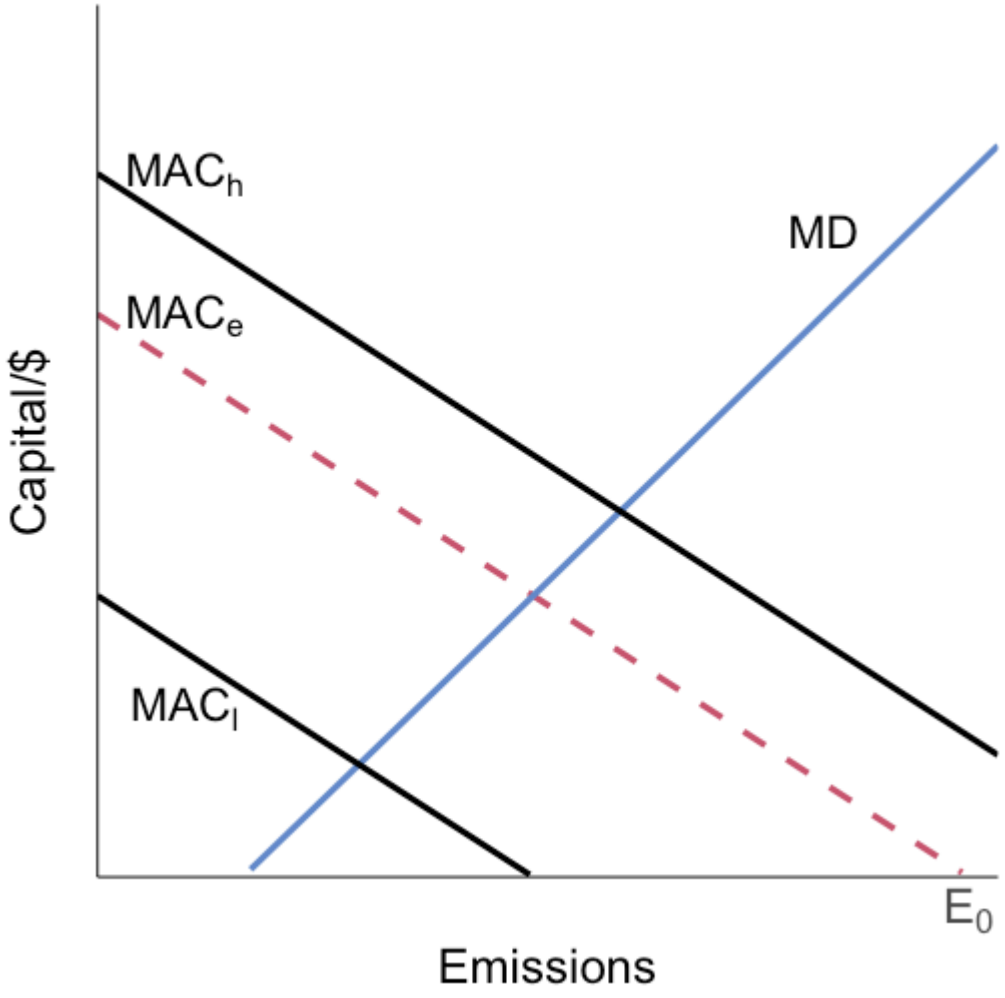
Policy under MAC uncertainty



Now suppose the regulator does not know the firms' **MAC curves**

The regulator's best estimate is MAC_e

Policy under MAC uncertainty

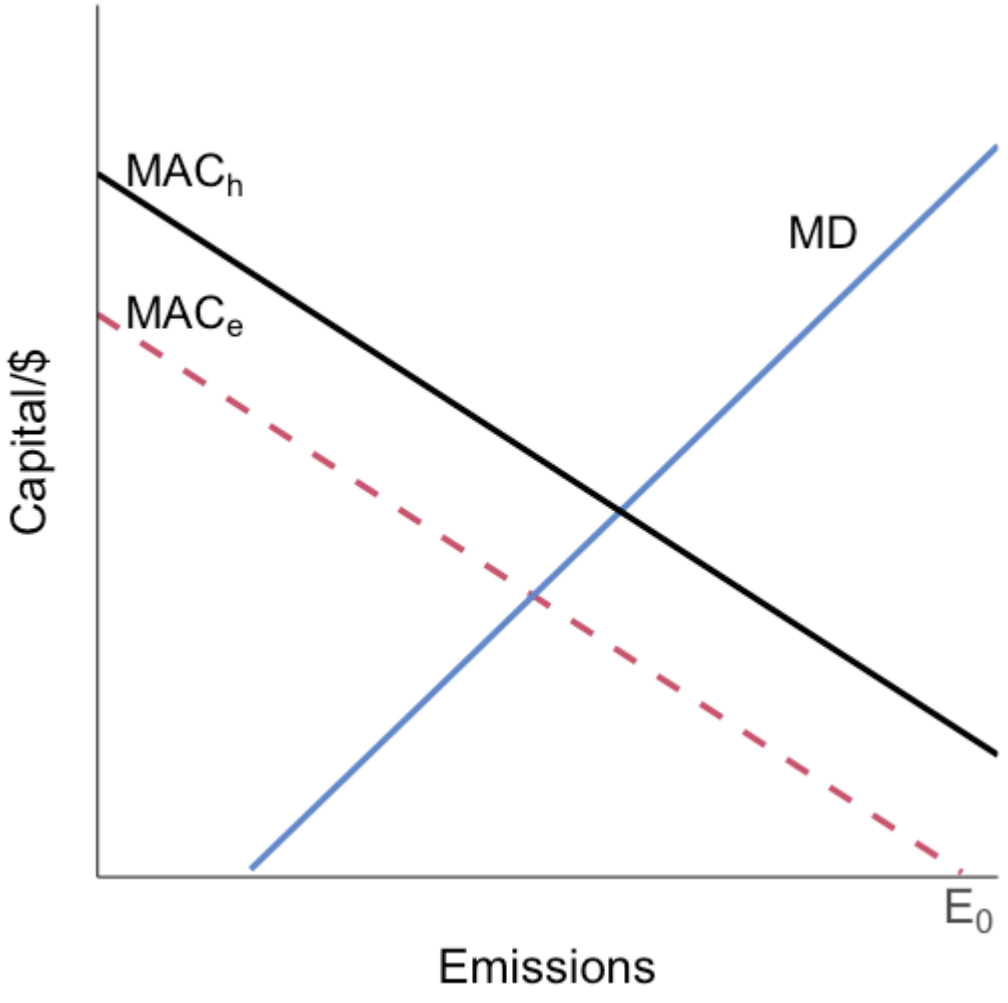


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The true MAC may instead be MAC_h or MAC_l

Policy under MAC uncertainty



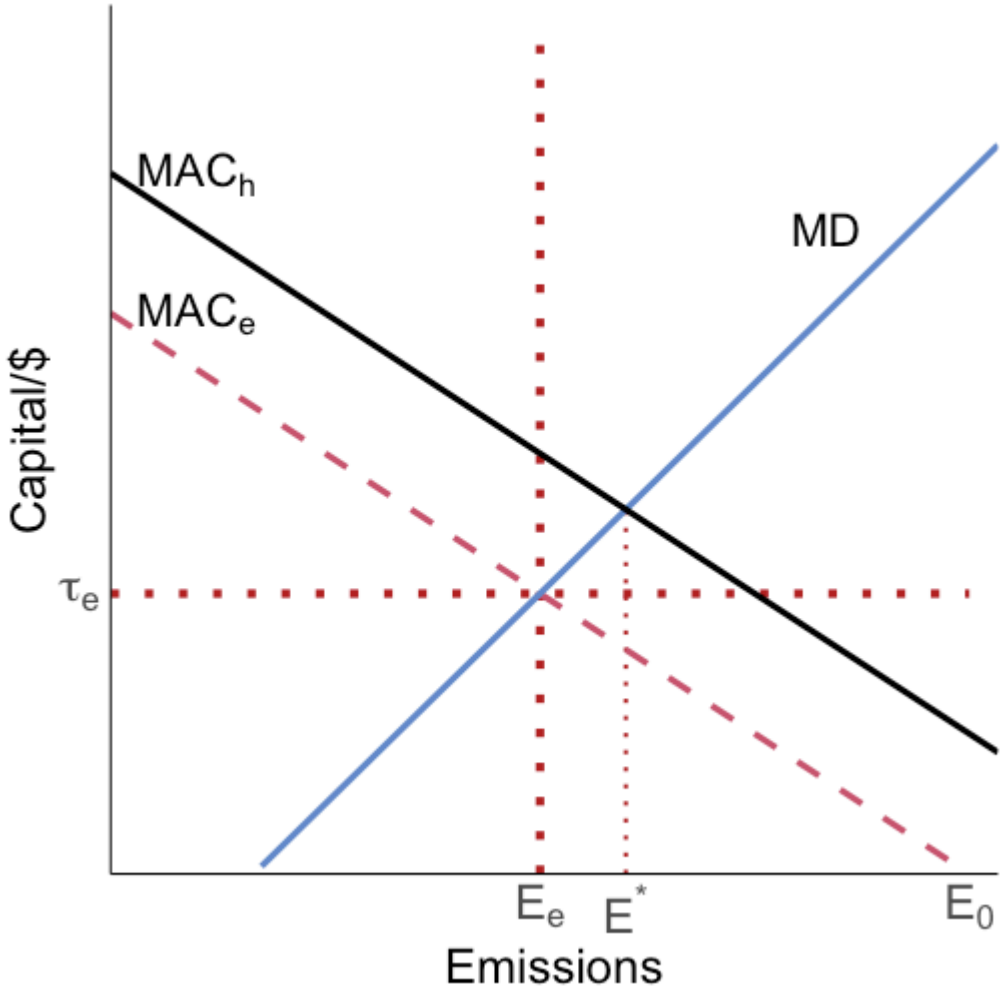
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Suppose the true MAC is MAC_h

Policy under MAC uncertainty

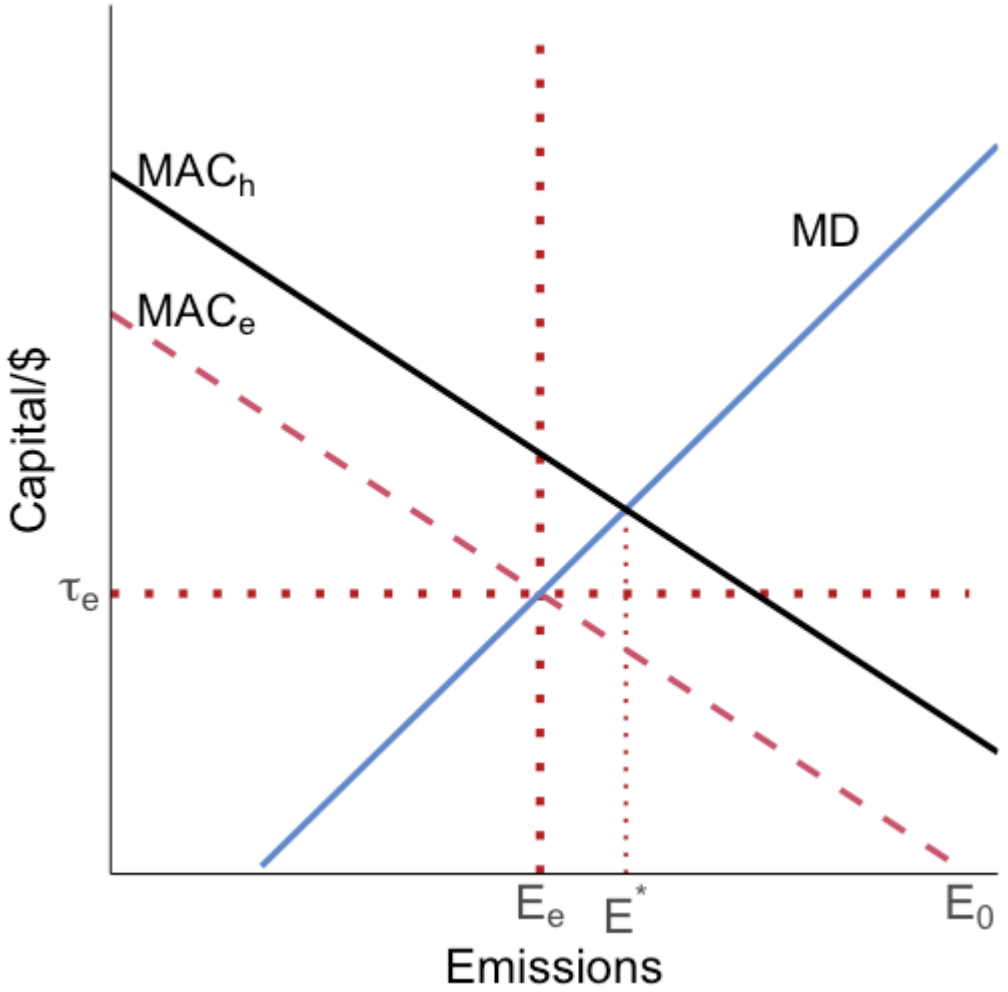


The regulator again sets a standard of E_e and a tax of τ_e

We know the efficient level of emissions is E^*

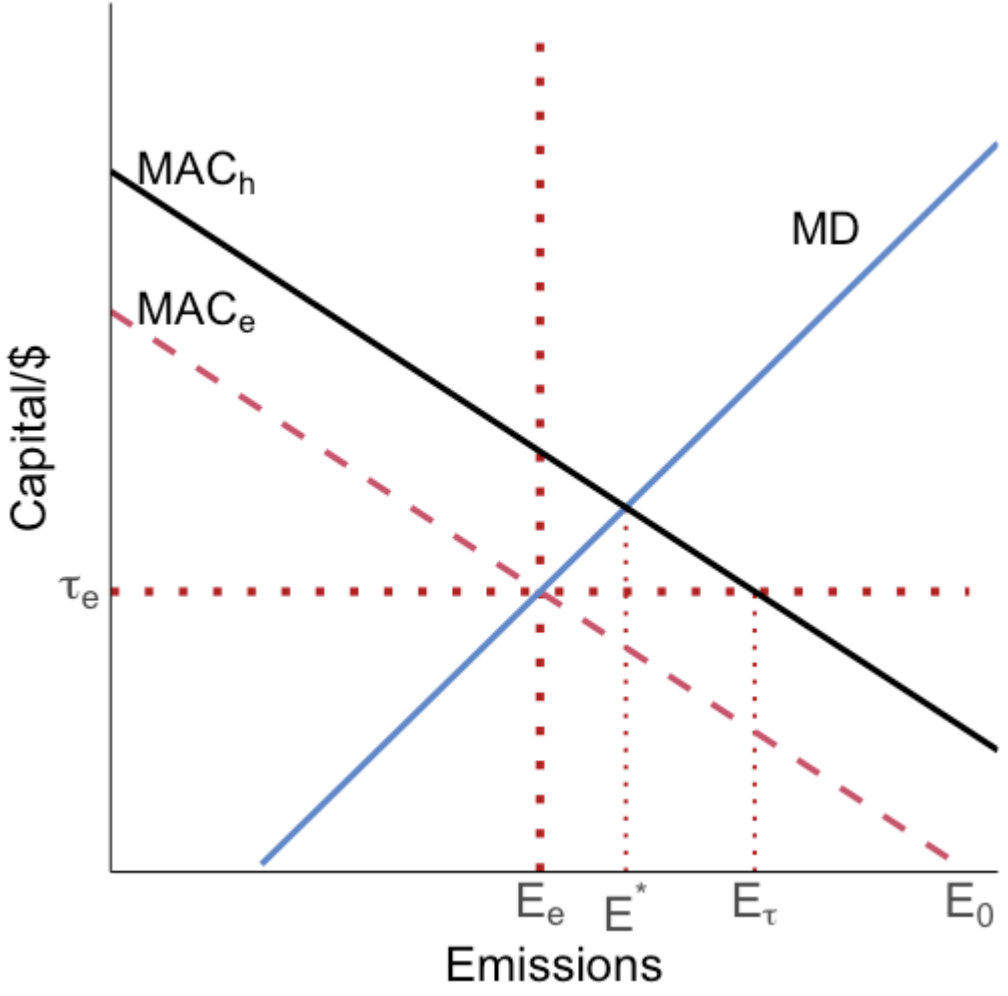
What level of emissions do we actually get?

Policy under MAC uncertainty



Under a standard we get E_e

Policy under MAC uncertainty



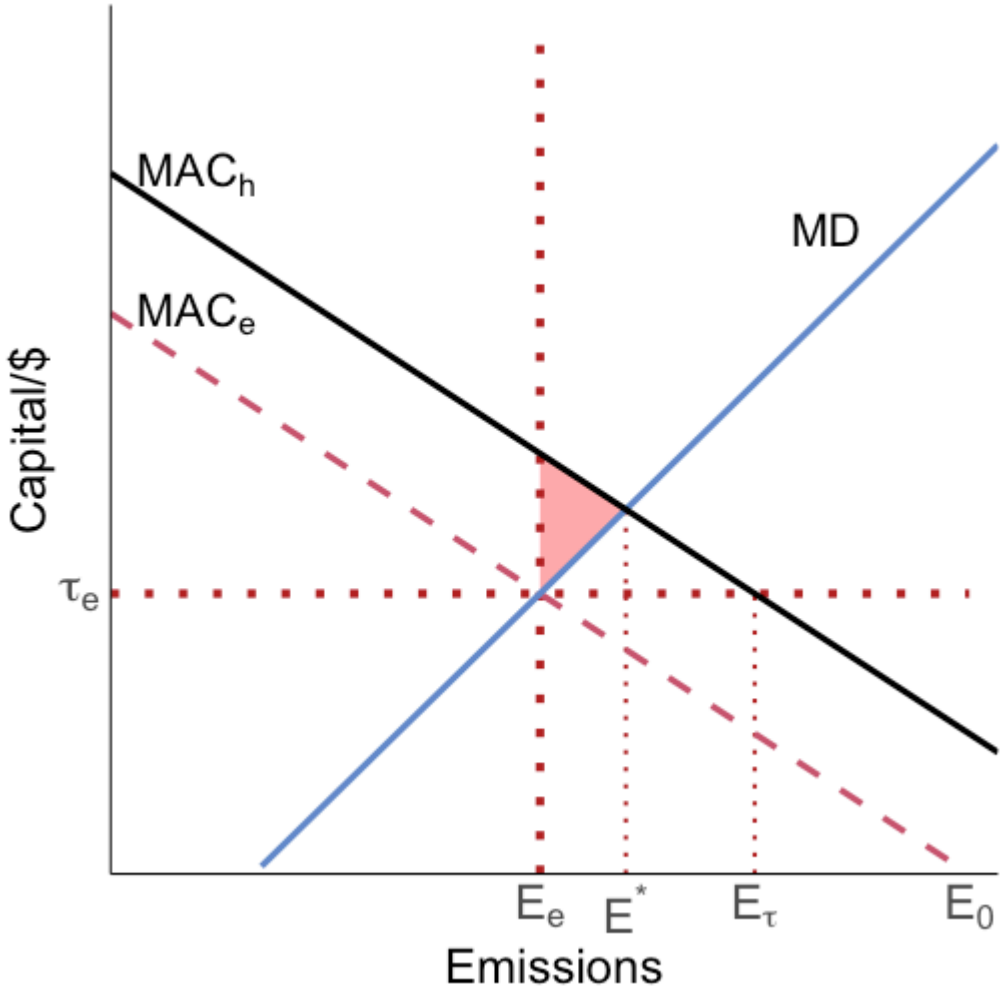
Under a standard we get E_e

Under a tax we get E_τ

Since we get different levels of emissions under the two policies, they may give us different levels of deadweight loss

Let's see

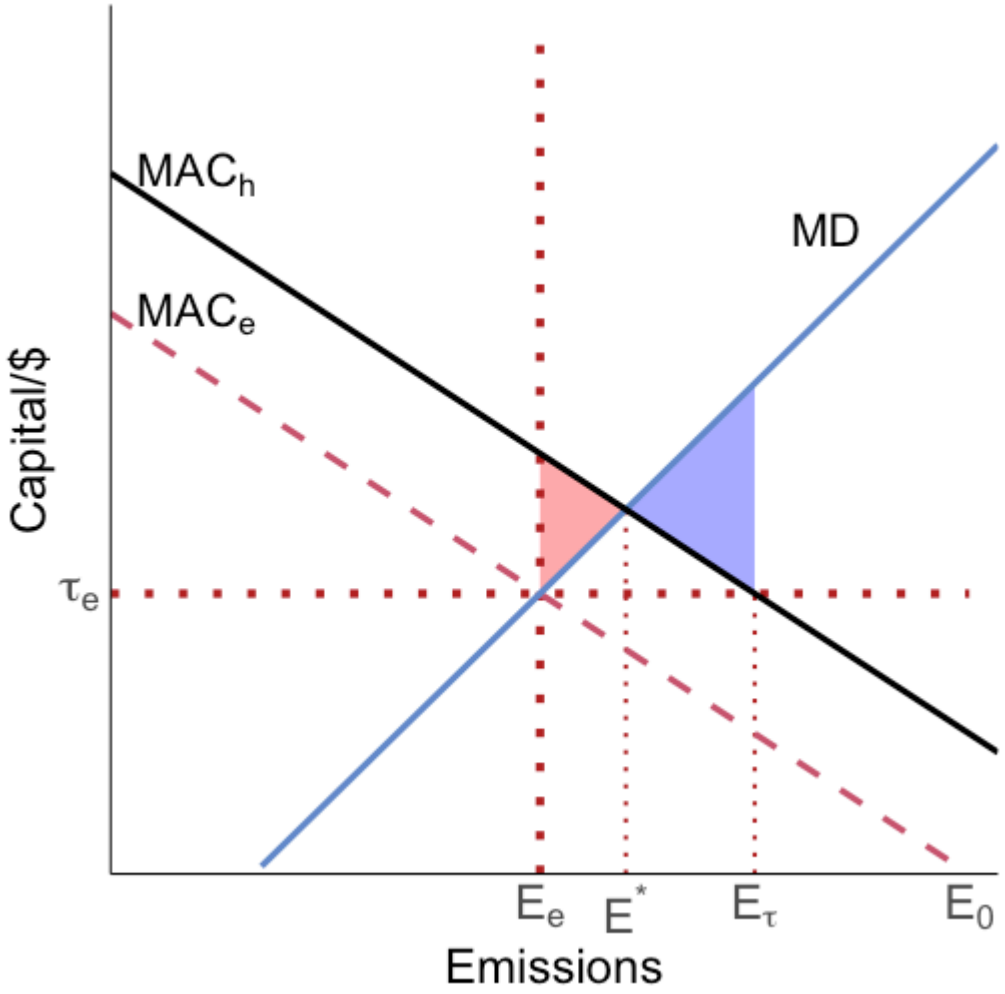
Policy under MAC uncertainty



The DWL under a standard is given by the **red** area

This is caused by **too little emissions**

Policy under MAC uncertainty



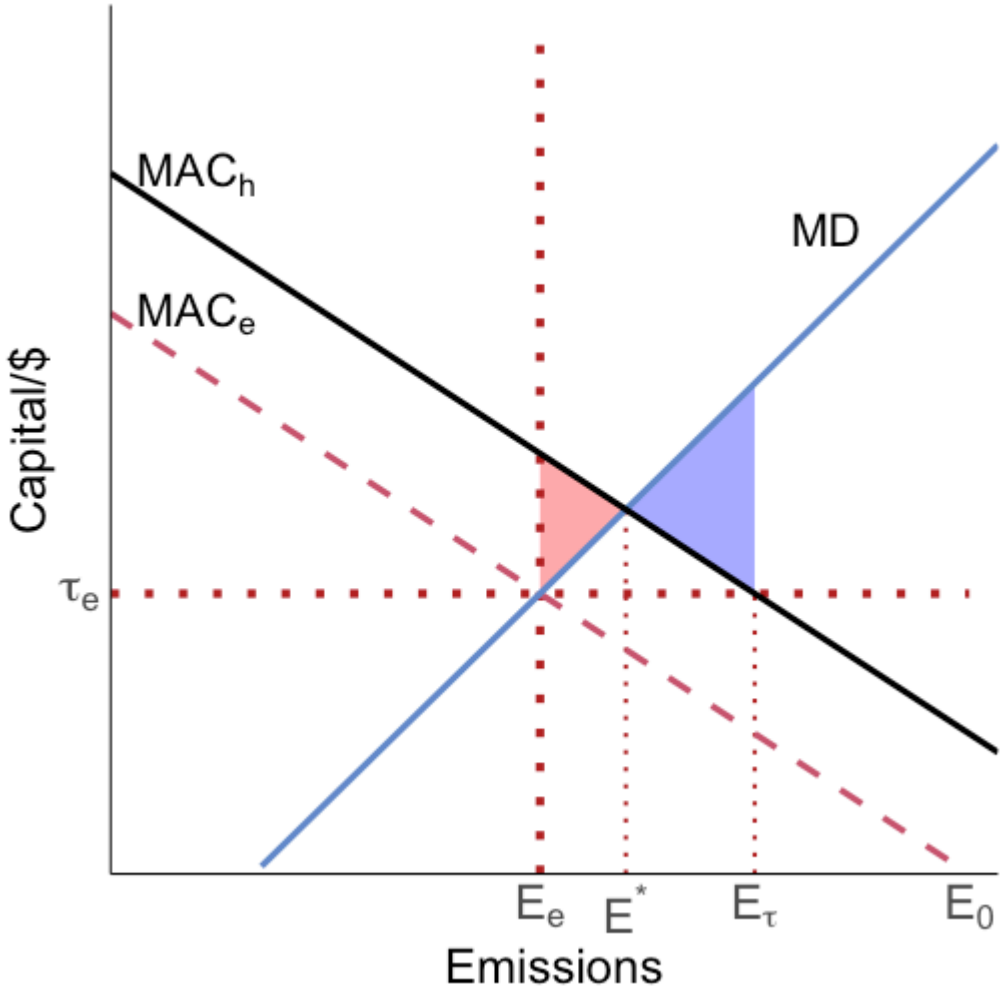
The DWL under a standard is given by the **red** area

This is caused by **too little emissions**

The DWL under a tax is given by the **blue** area

This is caused by **too many emissions**

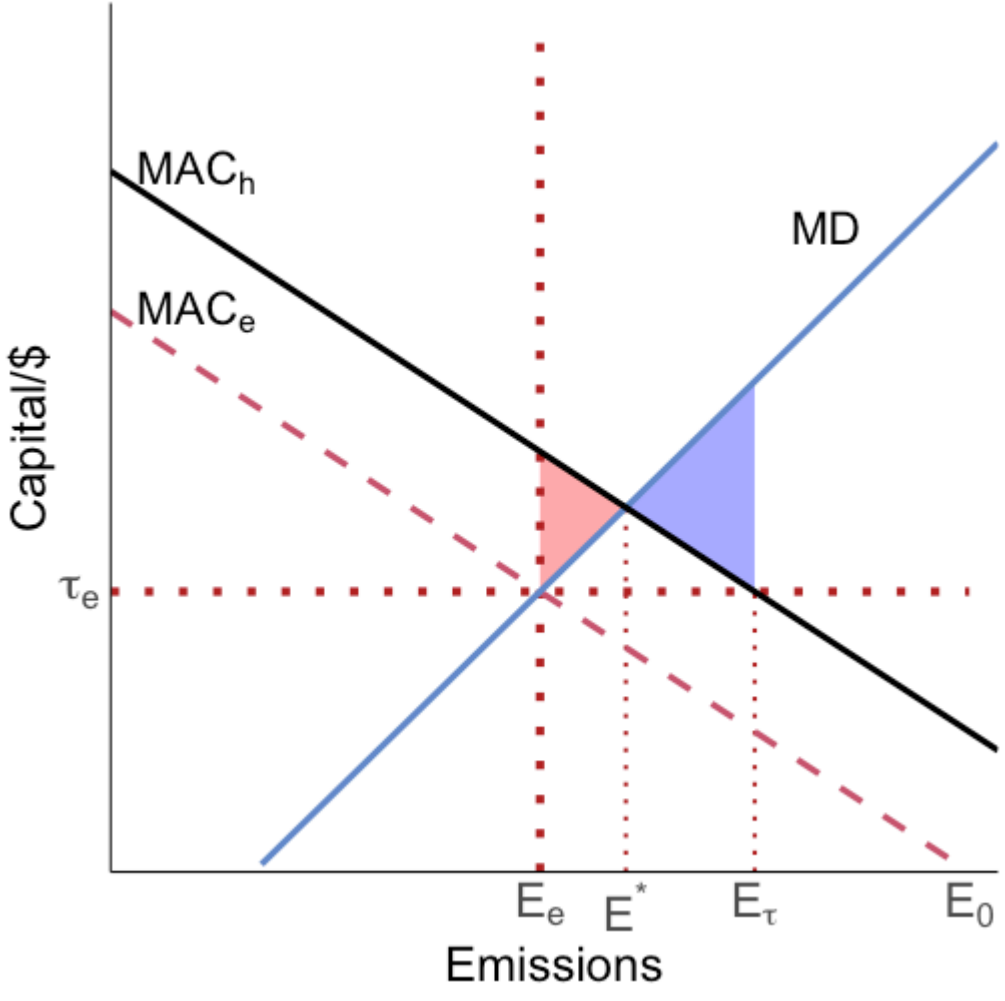
Policy under MAC uncertainty



The DWLs are similar-ish in size

Why do we get the DWLs?

Policy under MAC uncertainty

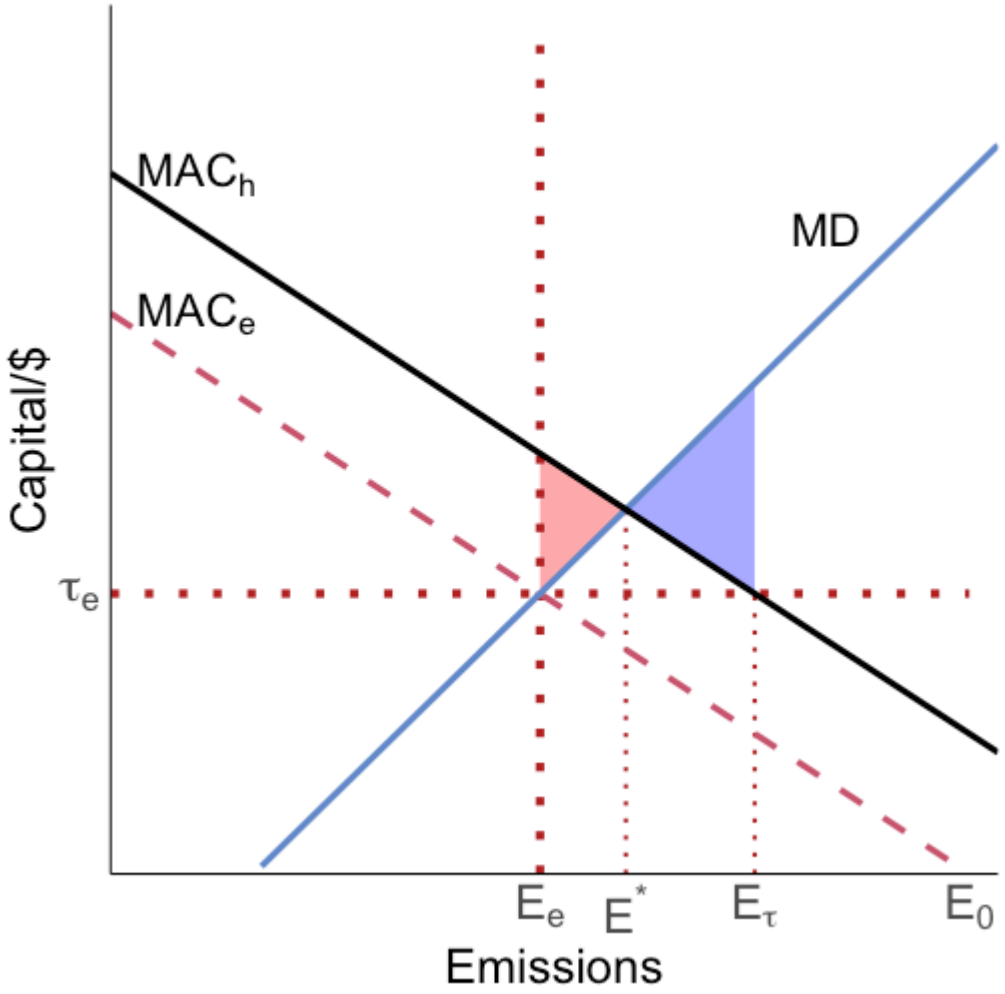


Standard: we under estimated the MAC and thought it'd be cheaper to abate than it actually was, so we set too stringent of a standard

Tax: we under estimated the MAC and thought it'd be cheaper to abate than it actually was, so we set too low of a tax

The same error leads to different qualitative outcomes

Policy under MAC uncertainty

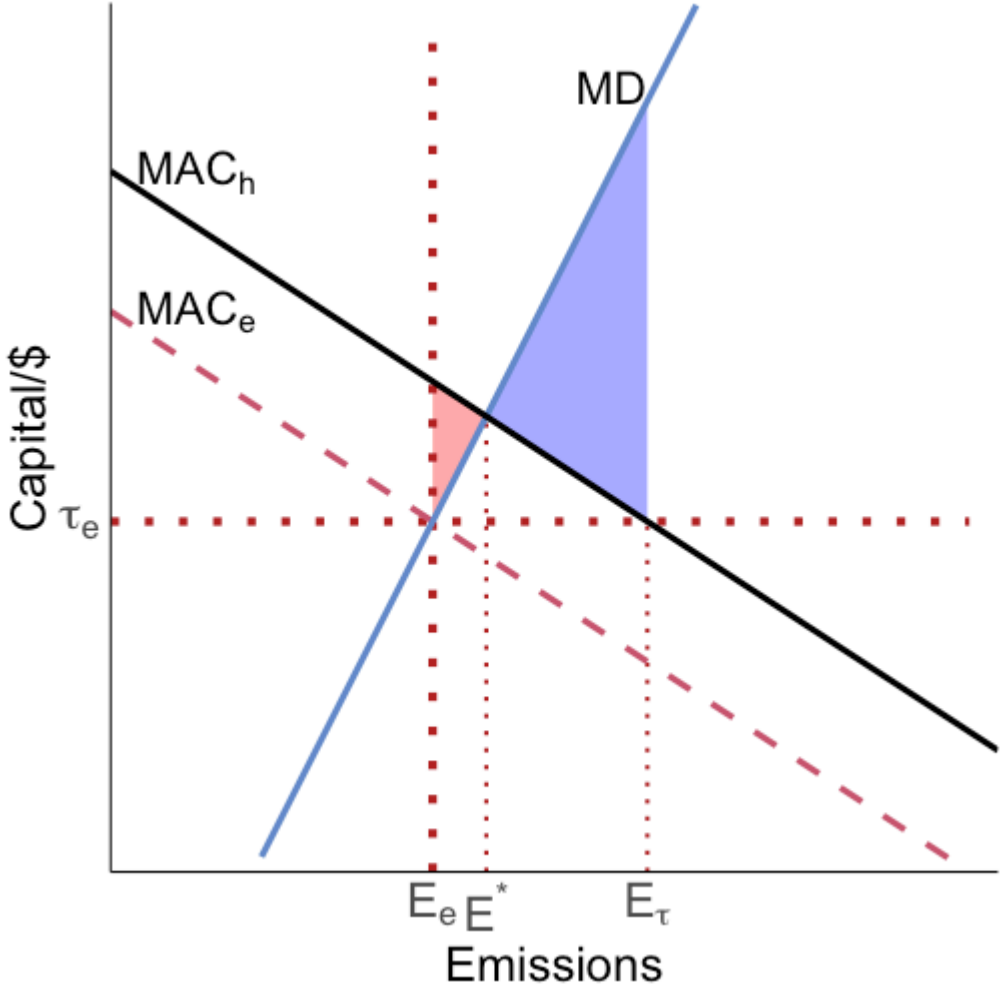


Which policy performs better depends on the relative slopes of the MAC and MD curves

Now consider an example where the MD curve is relatively **steep**

Try to solve for the policy choices and DWLs yourself

Policy under MAC uncertainty



If the MD curve is relatively steep a standard performs better, the **red** DWL is smaller than the **blue** DWL

Why?

Taxes allow firms to choose the quantity

If the MD is relatively steep, an error in the quantity of emissions is very costly

Steep MDs

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Any examples?

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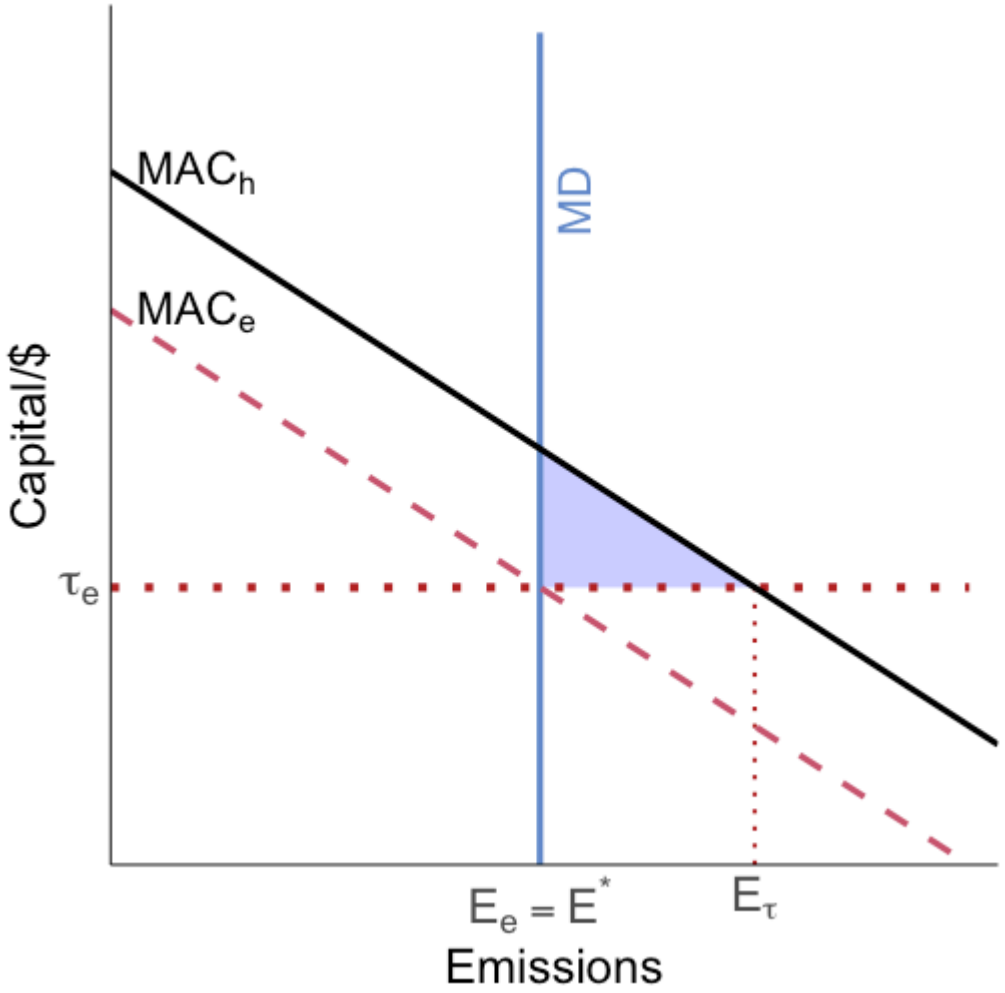
Any examples?

One would be if there exist **tipping points** in the climate system:

- Melt-back of Arctic summer sea ice
- Evidence that this is happening even without much additional warming
- Sea ice reflects light and cools polar regions
- This has big effects on air and ocean circulation

Collapse of Greenland ice sheet: could lead to a 7m rise in sea level

Policy under MAC uncertainty



If MD is **perfectly steep**, the efficient quantity is pinned down exactly

The standard still gives $E_e = E^*$, so there is **no DWL under the standard**

The tax still lets emissions move to E_τ when MAC is higher than expected

Perfectly steep MD and instrument choice

If MD is **perfectly steep**, damages spike at E^*, so we do not need to know MAC to identify the efficient quantity

A **standard** set at E^* pins down emissions at the efficient quantity

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The tax can cause **too much abatement**, or **too little abatement** and emissions beyond the damage threshold, potentially causing catastrophic damages

Tipping points

Collapse of western Antarctic ice sheet

- Potential for immediate sea level rise

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Die-back of Amazonian rainforest

- Species loss
- Effects on rainfall patterns in other parts of the world

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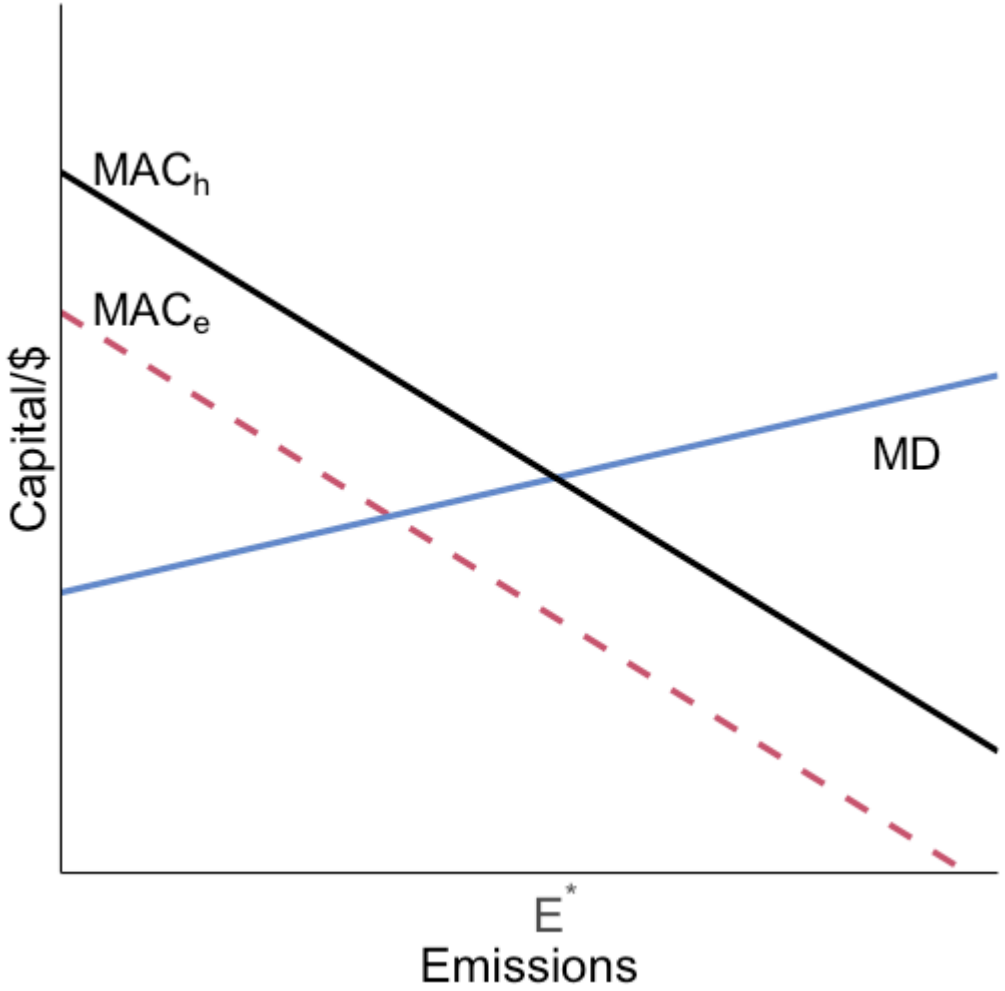
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Collapse of West African monsoon season and many others

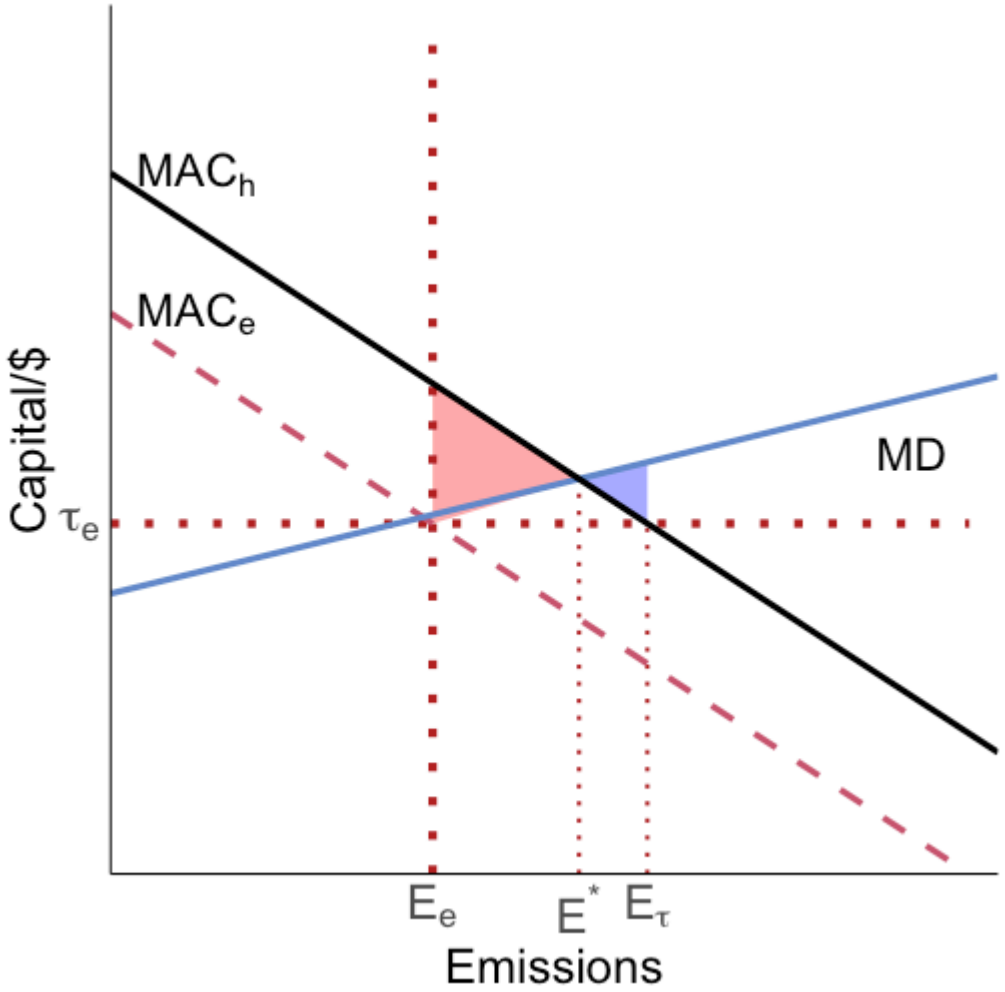
Policy under MAC uncertainty



Now what if MD is very flat?

Solve for the regulator's chosen policies and their associated DWL

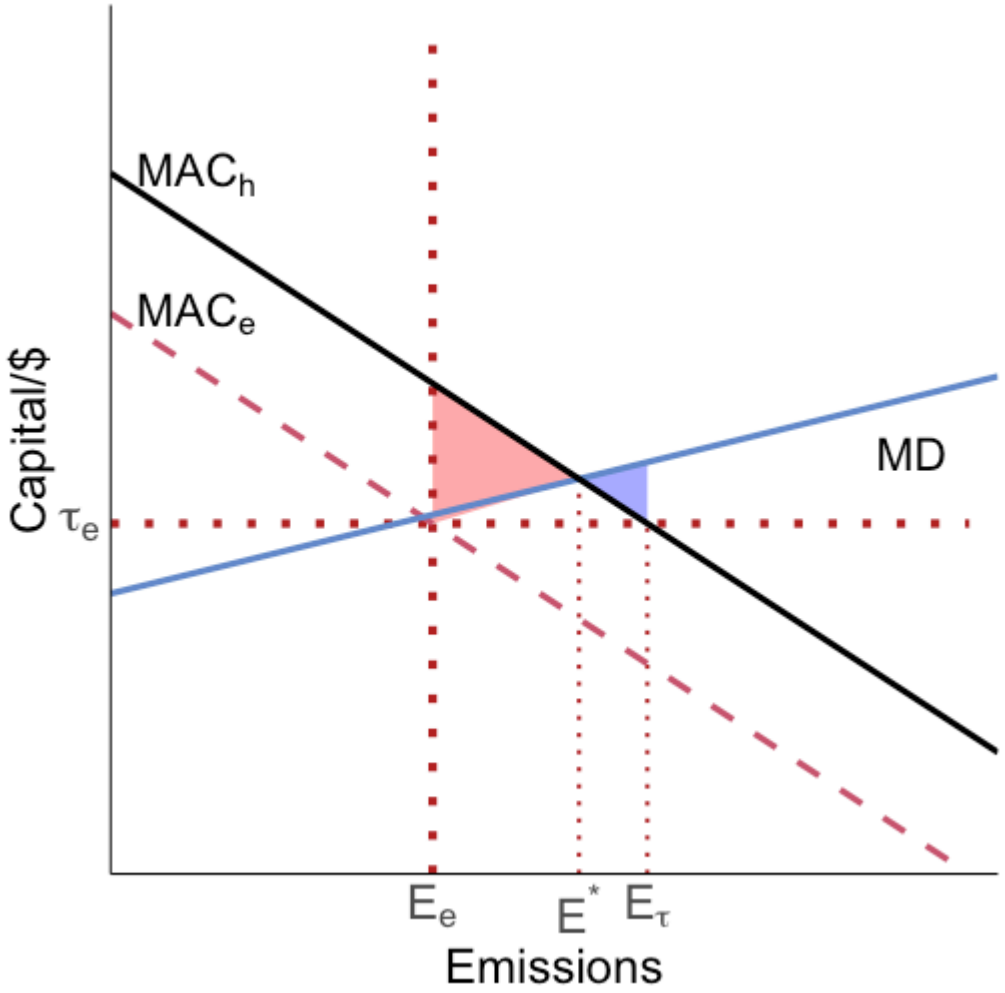
Policy under MAC uncertainty



If the MD curve is relatively flat a tax performs better, the **blue** DWL is smaller than the **red** DWL

Why?

Policy under MAC uncertainty

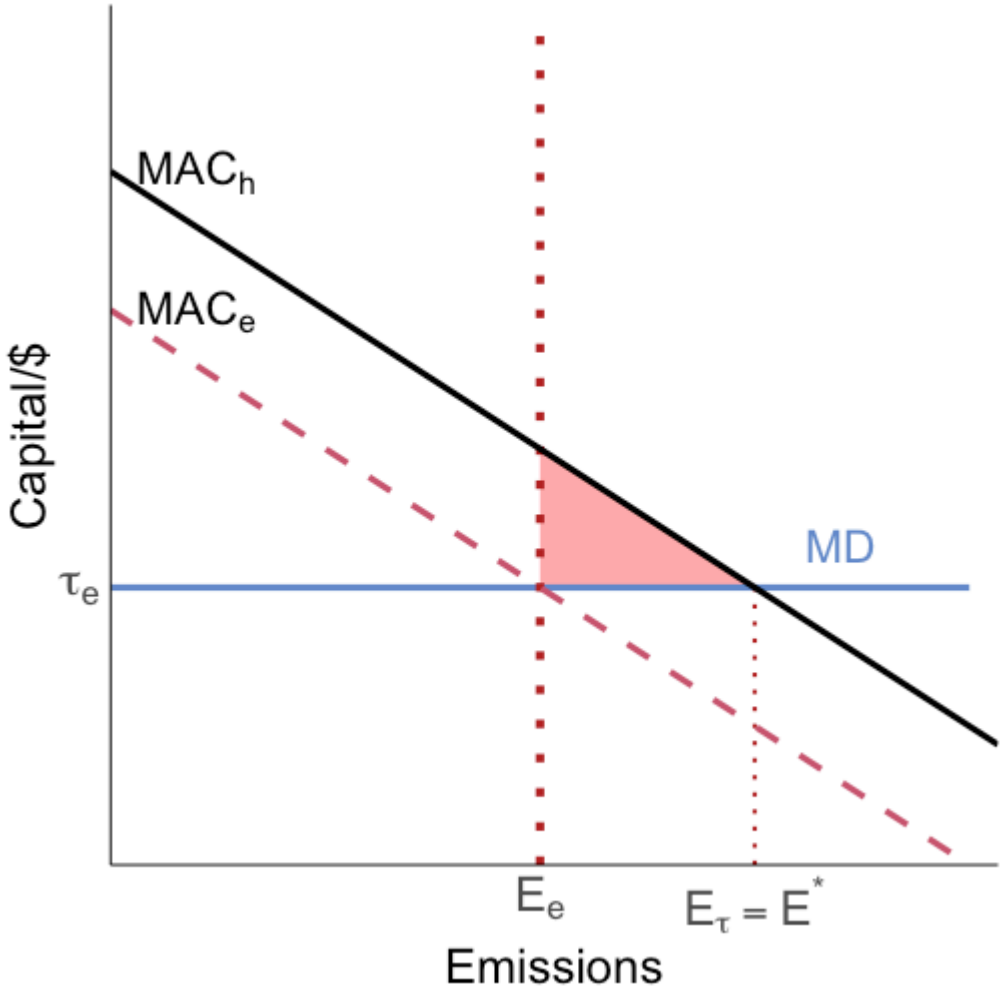


Marginal damages are relatively constant no matter the quantity while abatement costs can change rapidly

It's valuable to let the firm flexibly choose abatement

Taxes impose a constant incentive on the firm, closer representation of the MD

Policy under MAC uncertainty



If MD is **perfectly flat**, the efficient price is pinned down exactly

The tax still gives $\tau_e = MD$, so the firm chooses $E_\tau = E^*$

There is **no DWL under the tax**, while the standard still fixes the wrong quantity

Perfectly flat MD and instrument choice

If MD is **perfectly flat**, the marginal external cost of emissions is known and constant, so we do not need to know MAC to identify the efficient price

A **tax** set at $\tau = MD$ makes the firm pay that cost and choose $MAC = MD$, even though the regulator does not know MAC

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The implicit price can be **too high or too low** relative to MD, so the standard can cause **too much or too little abatement**

The Manne-Richel model

Two periods (current and future)

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Two possible states of nature in the future (“good” with probability (1-p) and “bad” with probability p)

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In the bad state, the climate is **very** sensitive to greenhouse gases; in the good state, it is less sensitive

The Manne-Richel model

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Two possible states of nature in the future ("good" with probability (1-p) and "bad" with probability p)

In the bad state, the climate is **very** sensitive to greenhouse gases; in the good state, it is less sensitive

Each state requires different abatement to avoid climate catastrophe

State	Probability	Cumulative Abatement Required
Good	1-p	0
Bad	p	1

The Manne-Richel model

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Since greenhouse gases are very long-lived, abatement activity is cumulative (doing abatement activity today is as good as doing abatement activity in the future)

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a_1: current abatement

a_2: future abatement

a_1+a_2: aggregate abatement

The Manne-Richel model

If we do a_1 units of abatement today in the future we need to do:

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1 - a_1 units with probability p (bad state)

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0 with probability 1-p (good state)

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Suppose abatement cost is given by: $C(a) = 0.5a^2$

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What is the regulator's problem?

The Manne-Richel model

$$\min_{a_1} E[TC] = \underbrace{\frac{1}{2}a_1^2}_{\text{current cost}} + (1-p) \times \underbrace{0}_{\text{good state cost}} + p \times \underbrace{\frac{1}{2}(1-a_1)^2}_{\text{bad state cost}}$$

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The first-order condition is:

$$\frac{dE[TC]}{da_1} = a_1^* - p(1-a_1^*) = 0$$

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$$\frac{dE[TC]}{da_1} = a_1^* - p(1-a_1^*) = 0$$

This gives us that:

$$a_1^* = \frac{p}{1+p}$$

The Manne-Richel model

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If a bad state is guaranteed (p = 1), we want $a_1^* = 1/2, a_2^* = 1/2$

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If a bad state is guaranteed (p = 1), we want $a_1^* = 1/2, a_2^* = 1/2$

You can think about this as basically buying insurance

The Manne-Richel model

It makes sense to do some abatement today, even if we are not sure there will be bad climate change, because the MAC is cheap for the first few units

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In marginal terms: we want to equate the MAC today with the *expected* MAC in the future so that on average, we can't reallocate abatement and be better off

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$$a_1^* = p(1 - a_1^*)$$